

Algebra II with Trigonometry

Chapter 6.5

Olympiad Solutions (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file:
[4533652_alg-2trig-h-chp-6-5-note.pdf](#)

Solutions

1. **Phase Shift Estimation:** Evaluate exactly:

$\arcsin(\sin 10) + \arccos(\cos 10)$. (Assume the angle 10 is in radians).

Solution: Let $A = \arcsin(\sin 10)$ and $B = \arccos(\cos 10)$. By definition, $y = \arcsin(\sin x)$ returns the unique angle $y \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ such that $\sin y = \sin x$. We know the trigonometric identity $\sin x = \sin(3\pi - x)$. Thus, $\sin 10 = \sin(3\pi - 10)$. Since $3\pi \approx 9.42$, the angle $3\pi - 10 \approx -0.58$ properly falls within the restricted range $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Therefore, $A = 3\pi - 10$.

Similarly, $y = \arccos(\cos x)$ returns the unique angle $y \in [0, \pi]$ such that $\cos y = \cos x$. Using the identity $\cos x = \cos(4\pi - x)$, we have $\cos 10 = \cos(4\pi - 10)$. Since $4\pi \approx 12.57$, the angle $4\pi - 10 \approx 2.57$ safely falls within the range $[0, \pi]$. Therefore, $B = 4\pi - 10$.

Summing the two evaluated terms:

$$A + B = (3\pi - 10) + (4\pi - 10) = 7\pi - 20$$

2. **Nested Invariant:** Find the exact value of $\tan(\arcsin(\cos(\arctan 2024)))$.

Solution: Let $\theta = \arctan 2024$. This means θ is an acute angle in $(0, \frac{\pi}{2})$ associated with a right triangle where the opposite side is 2024 and the adjacent side is 1, giving $\tan \theta = 2024$.

Next, let $\alpha = \arcsin(\cos \theta)$. Because θ is acute, $\cos \theta \in (0, 1)$, which means α is also an acute angle in $(0, \frac{\pi}{2})$. By definition, we have $\sin \alpha = \cos \theta$. Because α and θ are acute angles whose sine and cosine are equal, they must be complementary angles, meaning $\alpha = \frac{\pi}{2} - \theta$.

We are asked to evaluate $\tan(\alpha)$. Using the complementary angle identity:

$$\tan(\alpha) = \tan\left(\frac{\pi}{2} - \theta\right) = \cot \theta = \frac{1}{\tan \theta}$$

Given that $\tan \theta = 2024$, the exact value is: $\frac{1}{2024}$

3. Algebraic Optimization: Find the maximum possible value of the function $f(x) = (\arcsin x)^3 + (\arccos x)^3$ over its real domain.

Solution: The domain of $f(x)$ is $x \in [-1, 1]$. Let $u = \arcsin x$ and $v = \arccos x$. From trigonometric identities, we know that $u + v = \frac{\pi}{2}$ for all valid x .

We can rewrite the function as a sum of cubes:

$$\begin{aligned} f(x) &= u^3 + v^3 = (u + v)(u^2 - uv + v^2) \text{ Substitute } u + v = \frac{\pi}{2} \text{ and } \\ v &= \frac{\pi}{2} - u: \\ f(x) &= \frac{\pi}{2}((u + v)^2 - 3uv) = \frac{\pi}{2}\left(\frac{\pi^2}{4} - 3u\left(\frac{\pi}{2} - u\right)\right) = \frac{\pi}{2}\left(3u^2 - \frac{3\pi}{2}u + \frac{\pi^2}{4}\right) \end{aligned}$$

This forms an upward-opening parabola with respect to u . To maximize $f(x)$, we must maximize the quadratic on the interval $u \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (the range of $\arcsin x$). The vertex of the parabola is at $u = \frac{\frac{3\pi}{2}}{2(3)} = \frac{\pi}{4}$. The maximum over a closed interval occurs at the endpoint furthest from the vertex, which is $u = -\frac{\pi}{2}$ (corresponding to $x = -1$).

Substituting $u = -\frac{\pi}{2}$ (so $v = \pi$), we obtain:

$$f(-1) = \left(-\frac{\pi}{2}\right)^3 + \pi^3 = -\frac{\pi^3}{8} + \pi^3 = \frac{7\pi^3}{8}$$

4. Trigonometric Equation: Determine the unique positive real number x that satisfies the equation $\arcsin(x) + \arcsin(2x) = \arccos(x)$.

Solution: Using the complementary angle identity $\arcsin x + \arccos x = \frac{\pi}{2}$, we can replace $\arccos x$ and rewrite the equation as:

$$\arcsin 2x = \frac{\pi}{2} - 2 \arcsin x$$

Taking the sine of both sides yields: $\sin(\arcsin 2x) = \sin\left(\frac{\pi}{2} - 2 \arcsin x\right)$
 $2x = \cos(2 \arcsin x)$

Let $u = \arcsin x$, meaning $\sin u = x$. Using the double angle identity for cosine ($\cos 2u = 1 - 2 \sin^2 u$), the equation becomes: $2x = 1 - 2x^2$

Rearranging into a standard quadratic equation gives: $2x^2 + 2x - 1 = 0$

Applying the quadratic formula, we find: $x = \frac{-2 \pm \sqrt{4 - 4(2)(-1)}}{4} = \frac{-1 \pm \sqrt{3}}{2}$

Since we require the unique positive root, we choose $x = \frac{\sqrt{3}-1}{2}$. To ensure this solution is not extraneous, we verify the domain and range:

$2x = \sqrt{3} - 1 \approx 0.732 \leq 1$. Additionally, $\arcsin x \approx \arcsin(0.366) < \frac{\pi}{4}$, ensuring the right-hand side of our rearranged equation, $\frac{\pi}{2} - 2 \arcsin x$, remains strictly positive, logically matching the positive output of $\arcsin(2x)$.

Thus, the valid unique positive solution is $x = \frac{\sqrt{3}-1}{2}$.

5. Telescoping Arguments: Evaluate exactly, using purely mental arithmetic: $\arctan\left(\frac{1}{3}\right) + \arctan\left(\frac{1}{5}\right) + \arctan\left(\frac{1}{7}\right) + \arctan\left(\frac{1}{8}\right)$.

Solution: We leverage the inverse tangent addition formula:

$\arctan a + \arctan b = \arctan\left(\frac{a+b}{1-ab}\right)$. Pairing the terms strategically, we first combine $\arctan\left(\frac{1}{5}\right)$ and $\arctan\left(\frac{1}{8}\right)$:

$$\arctan\left(\frac{1}{5}\right) + \arctan\left(\frac{1}{8}\right) = \arctan\left(\frac{\frac{1}{5} + \frac{1}{8}}{1 - \frac{1}{5} \cdot \frac{1}{8}}\right) = \arctan\left(\frac{\frac{13}{40}}{\frac{39}{40}}\right) = \arctan\left(\frac{1}{3}\right)$$

Our overall expression simplifies elegantly: $2 \arctan\left(\frac{1}{3}\right) + \arctan\left(\frac{1}{7}\right)$

Applying the addition formula to double $\arctan\left(\frac{1}{3}\right)$:

$$2 \arctan\left(\frac{1}{3}\right) = \arctan\left(\frac{\frac{2}{3}}{1 - \left(\frac{1}{3}\right)^2}\right) = \arctan\left(\frac{\frac{2}{3}}{\frac{8}{9}}\right) = \arctan\left(\frac{3}{4}\right)$$

Finally, adding this result to the remaining term, $\arctan\left(\frac{1}{7}\right)$:

$$\arctan\left(\frac{3}{4}\right) + \arctan\left(\frac{1}{7}\right) = \arctan\left(\frac{\frac{3}{4} + \frac{1}{7}}{1 - \frac{3}{4} \cdot \frac{1}{7}}\right) = \arctan\left(\frac{\frac{25}{28}}{\frac{25}{28}}\right) = \arctan(1) = \frac{\pi}{4}$$

6. Periodic Cancellation Graphs: Determine the total combined length of all intervals within the domain $x \in [0, 100\pi]$ for which the equation $\arcsin(\sin x) = \arccos(\cos x)$ holds true.

Solution: Let y be the common value of both sides of the equation:

$y = \arcsin(\sin x) = \arccos(\cos x)$. By the definitions of the principal inverse trigonometric functions, the range of $\arcsin$ is $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and the range of $\arccos$ is $[0, \pi]$. For the two evaluated expressions to be equal, their result y must lie in their intersecting ranges, meaning $y \in \left[0, \frac{\pi}{2}\right]$.

When $y \in \left[0, \frac{\pi}{2}\right]$, the respective equations $\arcsin(\sin x) = y$ and $\arccos(\cos x) = y$ mandate that $\sin x = \sin y$ and $\cos x = \cos y$. This simultaneous condition uniquely determines the angular position of x up to full 2π rotations. In other words, $x \equiv y \pmod{2\pi}$. Because y itself sits in the first quadrant ($y \in [0, \pi/2]$), it immediately follows that the equality holds if and only if: $x \pmod{2\pi} \in \left[0, \frac{\pi}{2}\right]$

Thus, the original equation is satisfied continuously on closed intervals of the form $\left[2k\pi, 2k\pi + \frac{\pi}{2}\right]$ for integers k . The specified domain $[0, 100\pi]$ encompasses exactly 50 complete periods of 2π . Within each full period, the interval of strictly valid solutions possesses a length of $\frac{\pi}{2}$.

Therefore, the total combined length of these intervals is: $50 \times \frac{\pi}{2} = 25\pi$